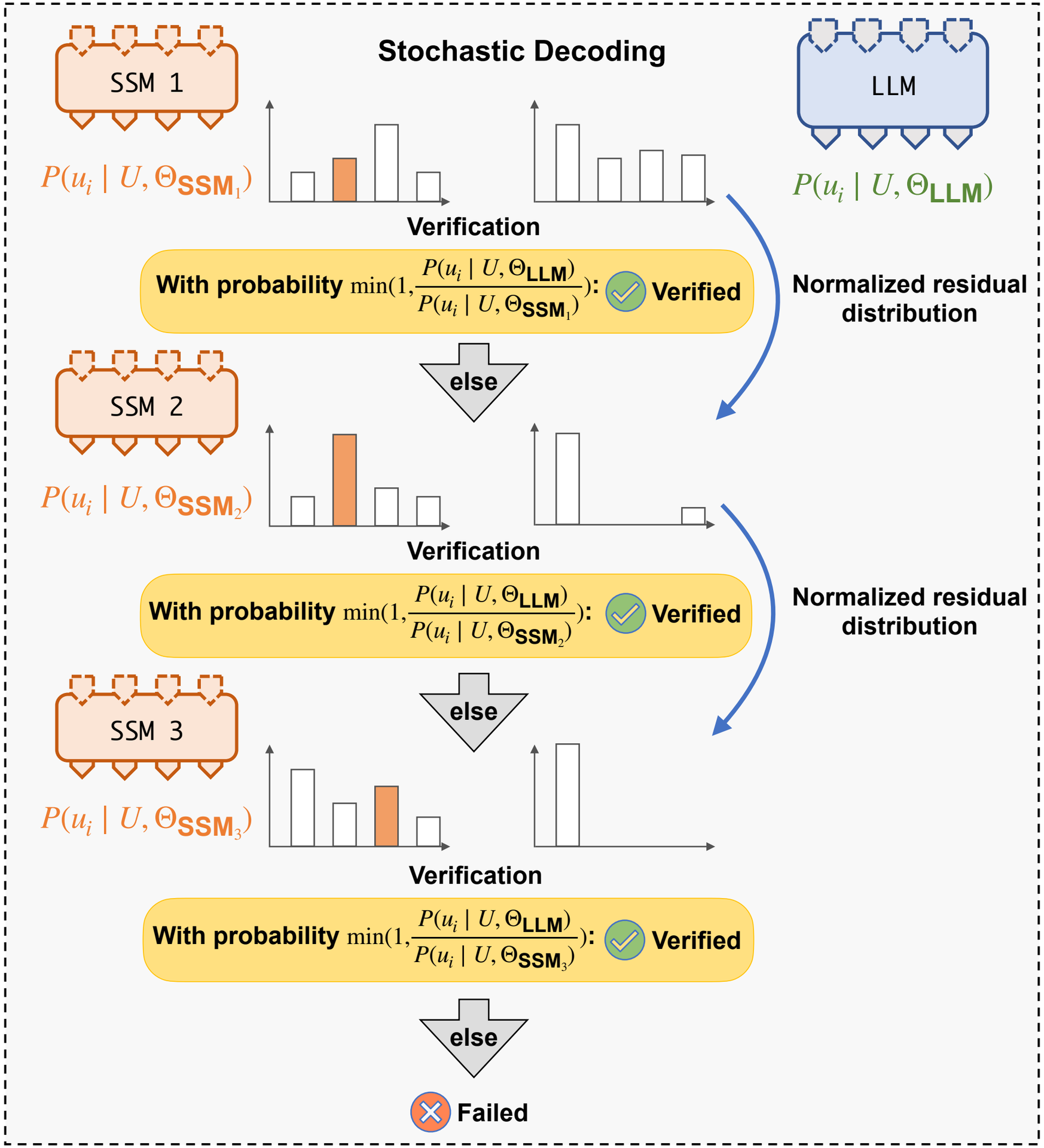

Stochastic Decoding
SSM 1
LLM
$P(u_i \mid U, \Theta_{\text{SSM}_1})$
$P(u_i \mid U, \Theta_{\text{LLM}})$
Verification
With probability $\min(1, \frac{P(u_i \mid U, \Theta_{\text{LLM}})}{P(u_i \mid U, \Theta_{\text{SSM}_1})})$: Verified
Normalized residual distribution
else
SSM 2
$P(u_i \mid U, \Theta_{\text{SSM}_2})$
Verification
With probability $\min(1, \frac{P(u_i \mid U, \Theta_{\text{LLM}})}{P(u_i \mid U, \Theta_{\text{SSM}_2})})$: Verified
Normalized residual distribution
else
SSM 3
$P(u_i \mid U, \Theta_{\text{SSM}_3})$
Verification
With probability $\min(1, \frac{P(u_i \mid U, \Theta_{\text{LLM}})}{P(u_i \mid U, \Theta_{\text{SSM}_3})})$: Verified
else
Failed